

ANAGRAPHICS

Born on November 21<sup>st</sup>, 1992 in Firenze (FI), Italy. Italian citizen. Pronouns: *he/him/his*.

EMPLOYMENT

|                     |                                                                                                                                                                  |                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Aug 2024 – present  | <b>Assistant Professor</b> in Topology,<br>Department of Mathematics, University of California Riverside, Riverside CA, USA.<br>Tenure track position.           | UC RIVERSIDE           |
| Aug 2023 – Jul 2024 | <b>Postdoctoral researcher</b> in Mathematics,<br>Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany.<br>Post-doc mentor: Anna Wienhard.     | MPI FOR MIS            |
| Aug 2020 – May 2023 | <b>Whyburn Research Associate and Lecturer</b> ,<br>Department of Mathematics, University of Virginia, Charlottesville VA, USA.<br>Post-doc mentor: Sara Maloni. | UNIVERSITY OF VIRGINIA |

EDUCATION

|                       |                                                                                                                                                                                                                       |                          |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Oct 2016 – Jul 2020   | <b>Ph.D.</b> in Mathematics,<br>Thesis title: <i>Constant curvature surfaces and volumes of convex co-compact hyperbolic manifolds</i> ,<br>Advisor: Jean-Marc Schlenker. (pdf file available <a href="#">here</a> .) | UNIVERSITÉ DU LUXEMBOURG |
| Sept 2014 – Sept 2016 | <b>M.Sc.</b> in Mathematics, $\frac{110}{110}$ , summa cum laude.<br>Thesis advisor: Riccardo Benedetti.                                                                                                              | UNIVERSITÀ DI PISA       |
| Sep 2011 – Sep 2014   | <b>B.Sc.</b> in Mathematics, $\frac{110}{110}$ , summa cum laude.<br>Thesis advisor: Paolo Lisca.                                                                                                                     | UNIVERSITÀ DI PISA       |

RESEARCH INTERESTS

(Higher) Teichmüller Theory, Anti-de Sitter geometry, Hyperbolic geometry, Kleinian groups,  
Low-dimensional topology, quantum Teichmüller space, quantum hyperbolic invariants.

PAPERS

|                        |                                                             |                                |
|------------------------|-------------------------------------------------------------|--------------------------------|
| Preprints<br>Published | <b>AWARDS AND GRANTS</b>                                    |                                |
| 2020                   | Excellent Thesis Award, year 2020.                          | UNIVERSITÉ DU LUXEMBOURG       |
| 2020                   | Graduate Internship grant.                                  | GEAR NETWORK                   |
| 2019                   | Travel grant for graduate students.                         | DPMA, UNIVERSITÉ DU LUXEMBOURG |
| 2017                   | Premio di Studio <i>Benedetto Sciarra</i> , anni 2015-2016. | SCUOLA NORMALE SUPERIORE, PISA |

LONG RESEARCH VISIT

|                     |                                      |                                   |
|---------------------|--------------------------------------|-----------------------------------|
| Jan 2019 – Mar 2019 | Supervised by Prof. Francis Bonahon. | UNIVERSITY OF SOUTHERN CALIFORNIA |
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INVITED TALKS

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## CONFERENCES AND WORKSHOPS

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|--------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------|
| May 23 <sup>rd</sup> , 2025                      | <i>Higher rank geometric structures, Higgs bundles and physics</i> , conference.                       | INSTITUT HENRI POINCARÉ         |
| Apr 28 <sup>th</sup> -May 2 <sup>nd</sup> , 2025 | <i>Leaning towards higher rank</i> , conference.                                                       | UNIVERSITÀ DI PISA              |
| Apr 15 <sup>th</sup> -19 <sup>th</sup> , 2024    | <i>Submanifolds of locally symmetric spaces: geometric, analytic and dynamical aspects</i> , workshop. | AUTRANS, FRANCE                 |
| Feb 21 <sup>st</sup> -23 <sup>rd</sup> , 2024    | <i>Mini-workshop on pseudo-Riemannian geometry</i> , workshop.                                         | UNIVERSITÀ DEGLI STUDI DI PAVIA |
| Jul 2023                                         | <i>Group actions and low-dimensional topology</i> , conference.                                        | ICMAT, EL BARCO DE ÁVILA, SPAIN |
| Jan 27 <sup>th</sup> , 2023                      | <i>Geometry, topology, and their applications</i> , online workshop.                                   | MPI FOR MiS, LEIPZIG, GERMANY   |
| Aug 22 <sup>nd</sup> -26 <sup>th</sup> , 2022    | <i>Minimal surfaces in symmetric spaces and Labourie's conjecture</i> , workshop.                      | AUTRANS, FRANCE                 |
| Jul 20 <sup>th</sup> , 2022                      | <i>AMS-EMS-SMF Meeting</i> , conference.                                                               | UNIVERSITÉ GRENOBLE ALPES       |
| Mar 12 <sup>th</sup> , 2022                      | <i>55<sup>th</sup> Spring Topology and Dynamics Conference</i> , conference.                           | BAYLOR UNIVERSITY               |
| Jun 2021                                         | <i>Nearly Carbon Neutral Geometric Topology Conference</i> , online conference.                        | ONLINE                          |
| Aug 2019                                         | <i>Combinatorial and algebraic aspects of geometric structures</i> , conference.                       | CHIANG MAI UNIVERSITY           |
| Dec 8 <sup>th</sup> , 2020                       | <i>Parole aux jeunes géomètres dynamiques</i> , online workshop.                                       | GDR PLATON                      |
| Sep 2018                                         | <i>Young Mathematicians Symposium of the Greater Region</i> , conference.                              | INRIA NANCY - GRAND EST         |
| May 2018                                         | <i>Structures on Surfaces</i> , workshop.                                                              | INRIA NANCY - GRAND EST         |

## MINI-COURSES

|           |                                |                |
|-----------|--------------------------------|----------------|
| Sept 2026 | <i>TBA</i> , Research program. | LMSI INSTITUTE |
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## SEMINARS

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|------------------------------|----------------------------------------------|----------------------------------|
| June 16 <sup>th</sup> , 2026 | Seminario di Geometria.                      | UNIVERSITÀ DEGLI STUDI DI TORINO |
| Mar 18 <sup>th</sup> , 2024  | Discrete groups and geometry.                | IHES                             |
| Mar 12 <sup>th</sup> , 2024  | Geometry and Topology Seminar.               | UNIVERSITÉ DU LUXEMBOURG         |
| Feb 15 <sup>th</sup> , 2024  | Séminaire de Théorie Spectrale et Géométrie. | INSTITUT FOURIER, GRENOBLE       |
| Oct 10 <sup>th</sup> , 2023  | Geometry Seminar.                            | MPI FOR MiS                      |
| Mar 6 <sup>th</sup> , 2023   | Dynamics Seminar.                            | UNIVERSITY OF WISCONSIN-MADISON  |
| Jan 13 <sup>th</sup> , 2023  | Colloquium talk.                             | UC RIVERSIDE                     |
| Nov 28 <sup>th</sup> , 2022  | Topology Seminar.                            | UT AUSTIN                        |
| Nov 18 <sup>th</sup> , 2022  | Geometry-Topology Seminar.                   | UC RIVERSIDE                     |
| Apr 4 <sup>th</sup> , 2022   | Geometry Seminar.                            | UNIVERSITY OF VIRGINIA           |
| Nov 29 <sup>th</sup> , 2021  | Geometry-Topology Seminar.                   | UNIVERSITY OF MARYLAND           |
| Jun 10 <sup>th</sup> , 2021  | Differential Geometry Seminar (online).      | HEIDELBERG UNIVERSITY            |
| May 11 <sup>th</sup> , 2021  | Geometry Seminar.                            | UNIVERSITY OF VIRGINIA           |
| Nov 3 <sup>rd</sup> , 2020   | Pangolin Seminar.                            | ONLINE                           |
| Oct 14 <sup>th</sup> , 2020  | Max Dehn Seminar.                            | UNIVERSITY OF UTAH               |
| Feb 12 <sup>th</sup> , 2020  | Topology Seminar.                            | TEXAS A&M                        |
| Jan 30 <sup>th</sup> , 2020  | Geometry & Topology Seminar.                 | BOSTON COLLEGE                   |
| Jan 28 <sup>th</sup> , 2020  | Geometry & Topology Seminar.                 | YALE UNIVERSITY                  |
| Jan 23 <sup>rd</sup> , 2020  | Differential Geometry Seminar.               | HEIDELBERG UNIVERSITY            |
| Dec 12 <sup>th</sup> , 2019  | Séminaire de Théorie Spectrale et Géométrie. | INSTITUT FOURIER, GRENOBLE       |
| Feb 19 <sup>th</sup> , 2019  | Geometry Seminar.                            | UNIVERSITY OF VIRGINIA           |

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TEACHING

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|             |                                                                          |              |
|-------------|--------------------------------------------------------------------------|--------------|
| Winter 2026 | MATH 011, Introduction to Discrete Structures, undergraduate course.     | UC RIVERSIDE |
|             | MATH 010A, Calculus of Several Variables, undergraduate course.          |              |
| Fall 2025   | MATH 259, Geometry and topology of surfaces, graduate course.            |              |
| Winter 2025 | MATH 010B, Calculus of Several Variables (part 2), undergraduate course. |              |
|             | MATH 009B, First-year Calculus (part 2), undergraduate course.           |              |

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| Fall 2024           | MATH 009B, First-year Calculus (part 2), undergraduate course.                                                               |                          |
| Jan 2024 – Feb 2024 | An introduction to anti-de Sitter geometry, mini-course.                                                                     | MPI FOR MİS              |
| Spring 2023         | MATH 8720, Hyperbolic Geometry, topics course.<br>MATH 5700, Introduction to Geometry, undergraduate course.                 | UNIVERSITY OF VIRGINIA   |
| Fall 2022           | MATH 3310-100, Basic Real Analysis, undergraduate course.                                                                    |                          |
| Spring 2022         | MATH 8720, Differential Geometry, graduate course.<br>MATH 5700, Introduction to Geometry, undergraduate course.             |                          |
| Fall 2021           | MATH 3310-100, Basic Real Analysis, undergraduate course.                                                                    |                          |
| Spring 2021         | MATH 3350-001, Applied Linear Algebra, undergraduate course.<br>MATH 3350-002, Applied Linear Algebra, undergraduate course. |                          |
| Fall 2020           | MATH 2310-101/201, Calculus III (TA), undergraduate course.                                                                  |                          |
| Fall 2019           | BASI-313, Analyse 1 (TA), undergraduate course.                                                                              | UNIVERSITÉ DU LUXEMBOURG |
| Fall 2018           | BASI-280, Analyse 1a & 1b (TA), undergraduate course.                                                                        |                          |
| Spring 2018         | BASI-10 Analyse 2a & 2b (TA), undergraduate course.                                                                          |                          |
| Fall 2017           | BASI-12, Analyse 3b (TA), undergraduate course.                                                                              |                          |

## MENTORING AND OUTREACH

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| Jan 2026–current | PhD Thesis,<br>Mentee: David Weed. Topic: <i>TBD</i> .                                                                                                                                                                                         | UC RIVERSIDE             |
| Jan-Jun 2024     | ENS Lyon Internship,<br>Mentee: Antoine Ablondi. Topic: <i>Pseudo-Riemannian hyperbolic geometry</i> .                                                                                                                                         | MPI MİS                  |
| Oct 2022         | Arbeitsgemeinschaft on Higher Teichmüller theory,<br>Mentee: Fernando Camacho Cadena. Topic: <i>Symplectic structures on the Hitchin component</i> .                                                                                           | OBERWOLFACH              |
| Oct 2022         | Geometry Lab,<br>Topic: <i>Hyperbolic geometry</i> . Organizer: Sara Maloni.                                                                                                                                                                   | UNIVERSITY OF VIRGINIA   |
| Summer 2021      | Research Experience for Undergraduates (REU),<br>Co-advisor: Sara Maloni. Students: K. Betts, T. Larsen, J. Utley, A. Vanis.<br>Project: <i>The tri-pants graph of the twice-punctured torus</i> . Preprint <a href="#">arXiv:2111.07136</a> . | UNIVERSITY OF VIRGINIA   |
| 2018-19          | Science Comic: Through the Looking Glass, Reflections of Science in Luxembourg,<br>Title: <i>Number theory in the past, present, and future</i> . Comic book available <a href="#">here</a> .                                                  | UNIVERSITÉ DU LUXEMBOURG |

## ORGANIZING

### SEMINARS

|                     |                                                                                                        |                                    |
|---------------------|--------------------------------------------------------------------------------------------------------|------------------------------------|
| 2025-2026           | Geometry & Topology seminar,<br>Co-organizers: Brian Collier.                                          | UC RIVERSIDE, RIVERSIDE CA, U.S.A. |
| 2025-2026           | Colloquium talks committee,<br>Co-organizers: Brian Collier, Tom Gannon, Siting Liu, Qi Zhang (chair). | UC RIVERSIDE, RIVERSIDE CA, U.S.A. |
| 2024-2025           | Colloquium talks committee,<br>Co-organizers: Yat Tin Chow, Brian Collier, Qian Zhang (chair).         | UC RIVERSIDE, RIVERSIDE CA, U.S.A. |
| Sep 2023 - Apr 2024 | Geometry Seminar,<br>Co-organizers: Samantha Fairchild, James Farre, Konstantinos Tsouvalas.           | MPI FOR MİS, LEIPZIG, GERMANY      |

### CONFERENCES AND WORKSHOPS

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| Oct 2024 | 2024 Fall Western Sectional Meeting, conference.<br>Session: <i>Geometry, topology and dynamics of character varieties</i> . Co-organizers: Brian Collier. | UC RIVERSIDE, RIVERSIDE, USA |
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| Jul 2024     | 2 <sup>nd</sup> AMS-UMI International Joint Meeting 2024, conference.<br>Session: Anosov representations and higher Teichmüller theory. Co-organizers: Sara Maloni, Andrea Tamburelli. | PALERMO, ITALY         |
| Aug-Sep 2023 | Metrics on higher Teichmüller spaces, workshop.<br>Co-organizers: Christian El Emam, Nathaniel Sagman.                                                                                 | FRÉJUS, FRANCE         |
| June 2022    | Nearly Carbon Neutral Geometric Topology Conference 2022, online conference.<br>Session: Anosov representations. Co-organizers: Gabriele Viaggi.                                       | ONLINE                 |
| Nov 2021     | Virginia Topology Conference, conference.<br>Co-organizers: Thomas Koberda, Sara Maloni, Mark Pengitore.                                                                               | UNIVERSITY OF VIRGINIA |

## PRESENTATIONS

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|-----------------------|----------------------------------------------------------------------------------------------------|--------------------------|
| Fall 2021             | Reading seminar,<br><i>An introduction to Higgs bundles</i> , with the Geometry group at UVA.      | UNIVERSITY OF VIRGINIA   |
| Fall 2020-Spring 2021 | Reading seminar,<br><i>Anosov representations</i> , with Sara Maloni and Alec Traaseth.            | UNIVERSITY OF VIRGINIA   |
| Fall 2018             | Reading seminar,<br><i>A hyperKähler extension of the Teichmüller space</i> , with Andrea Seppi    | UNIVERSITÉ DU LUXEMBOURG |
| Apr 2018              | Ph.D. Seminar,<br><i>About the McShane identity</i> .                                              | UNIVERSITÉ DU LUXEMBOURG |
| Sep 2017              | Ph.D. Away Days,<br><i>Classical applications of the H-Cobordism theorem</i> .                     | UNIVERSITÉ DU LUXEMBOURG |
| Sep 2017              | Geometry and Topology seminar,<br><i>Intertwining operators of the quantum Teichmüller space</i> . | UNIVERSITÉ DU LUXEMBOURG |
| Apr 2017              | Ph.D. Seminar,<br><i>A parametrization of complex projective structures on surfaces</i> .          | UNIVERSITÉ DU LUXEMBOURG |
| Sep 2015              | Seminari dei Baby Geometri,<br><i>Sulle Geometrie in dimensione 3</i> .                            | UNIVERSITÀ DI PISA       |

## SERVICE

Referee for the academic journals *Bulletin of the London Mathematical Society*, *Geometriae Dedicata*, *Geometric and Functional Analysis*, *Mathematische Annalen*, *Advances in Mathematics*, *International Mathematics Research Notices*, *IMRN*, *Journal of Differential Geometry*, *Journal of Topology*, *Algebraic & Geometric Topology*, *Geometry & Topology*, *Transactions of the American Mathematical Society*.

## SKILLS

|           |                                                            |
|-----------|------------------------------------------------------------|
| Speaking  | Italian (native), English (fluent), French (intermediate). |
| Languages | C Language, L <sup>A</sup> T <sub>E</sub> X, Matlab.       |

## REFERENCES

### RESEARCH

**Francis Bonahon**  
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University of Southern California  
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## TEACHING

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